

**Example 1.** Let  $p$  be a prime so  $q = 2p - 1$  is also prime.

Let  $a$  be an integer and prove and prove that  $a^q \equiv a \pmod{pq}$ .

**Proof.** First observe that  $q$  is distinct from  $p$  because, otherwise if  $q = p$  then

which is not possible, since  $1$  is not a prime.

Because  $p$  and  $q$  are distinct primes it follows that:

Hence the given congruence is equivalent to the system of congruences

The second congruence is an immediate consequence of:

Let's verify the first congruence. First let's consider the case that:

Next let's consider the case that:

Then the first congruence is equivalent, after multiplying by  $a^{-1} \pmod{p}$ , to

which we confirm below by using Fermat's Theorem and that  $q = 2p - 1$ .

□

A. **Pseudoprimes.** When we see the congruence:

$$2^{323} \equiv 314 \pmod{323}$$

we can immediately conclude that 323 is:

So we have applied the conclusion of Fermat's theorem to reveal compositeness.

But, this trick does not always work.

An integer  $n$  is an **absolute pseudoprime** if

1 it is composite and:

or equivalently if

2 it is composite and:

We shall see in a moment why these two definitions are equivalent.

For now, let's verify that  $561 = 3 \cdot 11 \cdot 17$  is an absolute pseudoprime.

Let  $a \perp 561$  be given, in which case we must show:

Since 3, 11, 17 are pairwise relatively prime, this is equivalent to the system

We confirm each using Fermat's Theorem.

Now let's see why the two definitions are equivalent. We need the following

Definition 1 implies Definition 2 because if  $a \perp n$  then we can

We show that Definition 2 implies Definition 1 using

**Lemma.** Definition 2 implies absolute pseudoprimes are:

**Proof.** Let  $n$  be an integer satisfying Definition 2 of an absolute pseudoprime.

Assume towards a contradiction that:

Then there exists:

## Lecture 16. B3 – Fermat's Little Theorem.

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By the Chinese remainder theorem, there exists an integer  $a$  such that:

Since  $a$  shares no prime factors with  $n$ , it follows that:

Therefore by Definition [2](#) we have:

Since  $p^2$  is a divisor of  $n$  we also have:

Since  $a \equiv p + 1 \pmod{p^2}$  this is equivalent to

$$(p + 1)^{n-1} \equiv$$

On the other hand, by the binomial theorem, and the fact that  $n \equiv$

$$(p + 1)^{n-1} \equiv$$

Together we have shown

which is a contradiction. So,  $n$  must have been square-free.

□

Now let  $n$  satisfy Definition [2](#) in which case the lemma implies

$$n =$$

and thus verifying Definition [1](#) is equivalent to verifying system

Let integer  $a$  be given. Let  $p$  be one of the prime factors of  $n$ . We need confirm

If  $p \mid a$  then:

If  $p \nmid a$  then by the Chinese remainder theorem, there exists integer  $a'$  such that

In particular,  $a' \perp n$ , and thus by Definition [2](#) we have

The choice of  $a \equiv p + 1 \pmod{p^2}$  is a bit tricky to explain. Underlying it is the fact that, as we will show, for such an  $a$ , then  $a^{p-1} \not\equiv 1 \pmod{p^2}$ , an observation that will become relevant much later in the course when we discuss **primitive roots**.

B. **Producing Pseudoprimes.** How might we come up with pseudoprimes?

**Theorem.** If

$n = p_1 p_2 \cdots p_r$  is a square-free product of at least two distinct primes

and if:

Then  $n$  is an absolute pseudoprime.

**Proof.** Because  $n$  is square-free, we can use [2](#) to show it is pseudoprime by

Let  $a \perp n$  be given and let  $p$  be a prime factor of  $n$ .

By assumption

Thus

□

We can use this fact to reconfirm that **561** is absolute pseudoprime.

Since  $561 = 3 \cdot 11 \cdot 17$  is composite and square-free, and

it follows that **561** is an absolute pseudoprime.